
UNIT 10 DECISION THEORY

Structure

- 10.0 Objectives
- 10.1 Introduction
- 10.2 Decision Making under Uncertainty
 - 10.2.1 Maximax Criterion
 - 10.2.2 Maximin Criterion
 - 10.2.3 The Minimax Regrets Criterion
 - 10.2.4 Laplace Criterion
- 10.3 Decision Making Under Risk
- 10.4 Decision Tree Analysis
- 10.5 Let Us Sum Up
- 10.6 Keywords
- 10.7 Suggested Further Readings / References
- 10.8 Answer to Check Your Progress
- 10.9 Unit End Questions
- 10.10 Assignment Questions
- 10.11 Solution to Assignment Questions

10.0 OBJECTIVES

After studying this unit, you should be able to,

- discuss the meaning and types of decision theory;
- explain the decision-making under uncertainty;
- describe the decision-making under risk situations; and
- explain the decision tree analysis.

10.1 INTRODUCTION

As an individual, we make many decisions every day. Sometimes these decisions are highly important and may have a long-term impact on our future. Decisions about the selection of a vehicle, purchase of a plot, renting a farm, investment in shares/stocks, etc, are all important decisions and one would like to make a correct choice out of the

available alternatives. Decision theory is defined as a body of several methods which facilitate the decision-maker to select wisely the best course of action from amongst several alternatives.

In this unit, we will study the various decision rules that are available to decision-makers under two different decision-making environments namely,

1. Decision-making under uncertainty, and
2. Decision-making under risk.

The decision under uncertainty are those problems in which each course of action can result in alternative outcomes, the probabilities of which cannot be assigned as a decision-maker does not know which outcomes can or will occur. For example, a farmer in a village may know the possible outcomes for the monsoon this year as heavy rainfall, moderate rainfall, and low rainfall but does not know the probability of its occurrence, as he may not have past information with him that he can use for prediction. For decision-making under risks, the decision-maker would be having the additional information in terms of probabilities or the chances of having heavy, moderate, and low rainfall based on past knowledge and records.

10.2 DECISION MAKING UNDER UNCERTAINTY

Under conditions of uncertainty, a farmer would know that the yield of the crop may be high, moderate, or low but would not have enough information to be able to use probabilities for the possible states that will occur. Further, these yields are dependent on the type of crops used by the farmer. Presently he has to decide amongst the three crops (A, B, and C), he knows that the possible yield (High, Moderate and Low) will yield the following payoff (profits in 1000's Rs) as shown in Table 10.1 below,

Table 10.1: Payoff matrix (profits in 1000's Rs)

	Alternate Strategies	States of Nature		
		High	Moderate	Low
Crops	A	Rs. 60	Rs. 42	Rs. – 10
	B	Rs. 85	Rs. 60	Rs. – 20
	C	Rs. 50	Rs. 25	Rs. – 12.5

The above table is called the payout table, we will try to help the farmer in decision making by demonstrating the use of the following decision rules:

1. The Maximax Criterion
2. The Maximin Criterion
3. The Minimax regret Criterion
4. Laplace's Criterion

10.2.1 Maximax Criterion

The maximax criterion is based on an optimistic rule assuming that the best situation will occur in the future. Considering the payoff in Table 10.2 described earlier we demonstrate how to use the maximax criterion.

Table 10.2. Payoff matrix

Rs. 60	Rs. 42	Rs. – 10
Rs. 85	Rs. 60	Rs. – 20
Rs. 50	Rs. 25	Rs. – 12.5

The following steps are to be followed,

Step 1: Select the maximum payoff for each alternative/strategy. We observe the result in following Table 10.3

Table 10.3: Payoff table

Crops	Payoff
A	Rs. 60
B	Rs. 85
C	Rs. 50

Step 2: Select the maximum of these maximum payoffs: we observe crop B with Rs. 85 as the maximum payoff. So we have selected the strategy with a maximum of maximums.

10.2.2 Maximin Criterion

The Maximin criterion is based on a pessimistic rule assuming that the worst situation is likely to occur in the future and we could like to maximize the profits. Using the payoff matrix given in Table 10.2 referred to earlier we explain the rule as follows:

Table 10.2: Payoff matrix

Rs. 60	Rs. 42	Rs. – 10
Rs. 85	Rs. 60	Rs. – 20
Rs. 50	Rs. 25	Rs. – 12.5

Step 1: Select the minimum payoff for each strategy from following Table 10.4.

Table 10.4: Payoff matrix

Crops	Payoff
A	Rs. – 10
B	Rs. – 20
C	Rs. – 12.5

Step 2: Select the maximum of the minimum payoff; we observe that crop A leads to the maximum payoff and hence using crop A as the strategy is recommended.

Note that in case, we have a payoff matrix for costs and not for the profit we will be using the minimax criterion wherein we will be minimizing the maximum costs likely to be incurred. The steps to be followed in such a situation will be as follows.

Step 1: Select the maximum cost from each strategy.

Step 2: Select the minimum of the maximum costs selected in step 1.

10.2.3 The Minimax Regrets Criterion

This technique calculates the opportunity loss or regrets for not taking the best decision under each state of nature and the use of the minimax regret criterion is discussed below for our payoff matrix shown in Table 10.2;

Table 10.2: Payoff matrix

Rs. 60	Rs. 42	Rs. – 10
Rs. 85	Rs. 60	Rs. – 20
Rs. 50	Rs. 25	Rs. – 12.5

Step 1: Develop a regret matrix from the given payoff matrix, on the following principle:

- When payoff represents profit
 $i^{\text{th}} \text{ regret} = (\text{maximum payoff} - i^{\text{th}} \text{ pay off}) \text{ for the } j^{\text{th}} \text{ event}$
- When payoff represents costs
 $i^{\text{th}} \text{ regret} = (\text{maximum payoff} - i^{\text{th}} \text{ pay off}) \text{ for the } j^{\text{th}} \text{ event}$

The payoff matrix for the problem we have selected for illustration represents a profit payoff matrix. Therefore the regret Table 10.5 is prepared as follows;

Table 10.5: Profit payoff matrix

Alternative	Pay off amounts			Regret Pay off amounts			Maximum Regret
	High	Medium	Low	High	Medium	Low	
A	Rs. 60	Rs. 42	Rs. - 10	25	18	0	25
B	Rs. 85	Rs. 60	Rs. - 20	0	0	0	10
C	Rs. 50	Rs. 25	Rs. – 12.5	35	35	2.5	35

From the regret payoff matrix, the minimum regret is observed for crops B (Regret = 10) and therefore the selection of Crop B will be recommended.

10.2.4 Laplace Criterion

Also known as the equally likely decision criterion, this rule assumes that the states of nature (Low, Moderate, and High yield) will occur with equal probabilities. The expected value for each alternative is then calculated. The calculations for farmers' payoff matrix are shown below;

$$\text{Crop A} = \frac{1}{3} (60) + \frac{1}{3} (42) + \frac{1}{3} (-10) = 30.66$$

$$\text{Crop B} = \frac{1}{3} (85) + \frac{1}{3} (60) + \frac{1}{3} (-20) = 41.66$$

$$\text{Crop C} = \frac{1}{3} (50) + \frac{1}{3} (25) + \frac{1}{3} (-12.5) = 20.83$$

It is observed that crop B has the highest expected value and hence will be selected by the farmer.

Once we have understood the different rules available for decision-makers under uncertain environments, the next logical question is which one is the best technique to be used in practice, as it is observed that the different techniques may produce different results (selection of alternatives). I would like to mention here that the use of a particular technique depends on the decision maker's style, risk-taking ability, past behaviour, whether he knows the rule adopted by his competitor, etc. Therefore it is expected that different decision-makers may deploy different rules to take the decision in a given situation.

Check Your Progress 10.1

Note:

1. The decision taken under uncertain conditions results in alternative outcomes
 - a. True
 - b. False
2. The decision taken under risk needs additional information.
 - a. True
 - b. False
3. The maximax criterion is based on _____ rule assuming that the best situation will occur in the future.
4. The Maximin criterion is based on _____ rule assuming that the worst situation is likely to occur in the future.

5. The Minimax Regrets Criterion technique calculates loss and profit.
 - a. True
 - b. False
6. In the Laplace Criterion technique, the outcome occurs with equal probabilities.
 - a. True
 - b. False
7. Which is the best technique to be used in practice?
 - a. Maximax
 - b. Maximin
 - c. Minimax
 - d. Laplace Criterion
 - e. All the above
8. The maximin criteria is a (n) _____ criteria.
 - a. Optimistic
 - b. Neutral
 - c. Pessimistic
 - d. None of the above
9. Table 10.6 indicates 12 possible payoffs for the manufacturing company's expansion decision.

Table 10.6: Payoff matrix

Alternatives	<i>States of nature (product demand)</i>			
	<i>High</i>	<i>Moderate</i>	<i>Low</i>	<i>Nil</i>
Expand	Rs. 50,000	Rs. 25,000	Rs. – 25,000	Rs. – 25,000
Contract	Rs. 70,000	Rs. 30,000	Rs. – 40,000	Rs. – 80,000
Subcontract	Rs. 30,000	Rs. 15,000	Rs. – 1,000	Rs. – 10,000

Select optimum decision criterion using Maximax criterion

10. A company has to choose one of the three types of biscuits: Creams, Coconut, and Glucose, Sales expected during next year are highly uncertain. The

marketing department estimates the profits considering manufacturing cost, promotional efforts, distribution setup, etc. as given in Table 10.7 below.

Table 10.7: Payoff matrix

Type of Biscuits	<i>Profits on the estimated level of sales (in lakhs) for quantities</i>		
	5,000	10,000	20,000
Cream (C)	15	25	45
Coconut (Co)	20	55	65
Glucose (G)	25	40	70

Select optimum decision criterion using Maximin criterion

10.3 DECISION MAKING UNDER RISK

As mentioned in the introductory section of this unit, this risk environment assumes that the decision-maker has sufficient knowledge about the different states of nature based on which he can assign the probability of occurrences for them. In our illustration example, the farmer is of the view that a high yield is likely to occur with a probability of 0.65, a moderate yield with a probability of 0.20, and a low yield with a probability of 0.15. We can then compute the expected value for each alternative as shown in Table 10.8.

Table 10.8: Payoff matrix

Alternate Strategies	States of Nature			Expected Value
	High yield	Moderate yield	Low yield	
<i>Probabilities</i>	0.65	0.20	0.15	
Crop A	Rs. 60	Rs. 42	Rs. – 10	45.90
Crop B	Rs. 85	Rs. 60	Rs. – 20	64.25
Crop C	Rs. 50	Rs. 25	Rs. – 12.5	35.62

Expected value for crop A = $0.65 \times \text{Rs. } 60 + 0.20 \times \text{Rs. } 42 + 0.15 \times \text{Rs. } (-10) = 45.90$

Expected value for crop B = $0.65 \times \text{Rs. } 85 + 0.20 \times \text{Rs. } 60 + 0.15 \times \text{Rs. } (-20) = 64.25$

Expected value for crop C = $0.65 \times \text{Rs. } 50 + 0.20 \times \text{Rs. } 25 + 0.15 \times \text{Rs. } (-12.5) = 35.62$

The farmer should now opt for crop B to gain a higher expected value.

Now let us consider a typical situation where decisions are taken daily. Naresh is a newspaper vendor who sells newspapers in a small complex. He buys the newspaper from a local dealer at Rs. 1 per unit and sells it for Rs. 1.2 per paper. Any unsold paper at the end of the day reduces its value to Rs. 0.30 and is sold to a scrap dealer. Naresh's demand fluctuated between 60 to 65 customers a day. Further, he loses customers if he

does not have sufficient stock. Naresh has been maintaining data on his customers. His data reveals that the newspaper demand in the past 300 days was as given in Table 10.9.

Table 10.9: Demand Table for 300 days.

Demand in units	60	61	62	63	64	65
No. of days	45	60	90	60	30	15
Probability of demand (i.e. no. of days / 300)	0.15	0.20	0.30	0.20	0.10	0.05

The probability that the demand may be 60, 61, 62, and so on can be computed by considering the relative frequency. The following payoff Table 10.10 is constructed.

Table 10.10: Payoff matrix

Stocking Policy	60	61	62	63	64	65
Probability of demand	0.15	0.20	0.30	0.20	0.10	0.05
60	12	12	12	12	12	12
61	11.1	12.2	12.2	12.2	12.2	12.2
62	10.2	11.3	12.4	12.4	12.4	12.4
63	9.3	10.4	11.5	12.6	12.6	12.6
64	8.4	9.5	10.6	11.7	12.8	12.8
65	7.5	8.6	9.7	10.8	11.9	13.0

In the above payoff table it is observed that, as the demand varies between 60 to 65, Naresh's profit also varies depending on how many units he has stocked for that day. It can be seen that If Naresh stocks 63 units and the demand for that day turns out to be 63 units, the total profit earned by him will be;

$$63 \text{ units} \times 0.20 \text{ (profit margin)} = 12.6$$

However, if the demand that day is below his stocking level of 63 units and only 60 units are sold then his profit will be;

$$(\text{profit from the sold unit}) - \text{loss from unsold units (i.e. Rs. 0.90 per unit)}$$

$$= (60 \text{ units} \times 0.2 \text{ profit/unit}) - (3 \text{ units} \times 0.90 / \text{unit})$$

$$= \text{Rs. } 12 - \text{Rs. } 2.7$$

$$= \text{Rs. } 9.3$$

If Naresh stocks 64 units and the demand for that day is 65 he will be able to make a maximum profit of Rs.12.8 as all the units are sold. Table 10.10 is also referred to as the conditional table.

The conditional table (refer to Table 10.10) for this problem tells how much profits Naresh will be able to make under different stocking and demand conditions, but it

does not tell how much Naresh is required to stock. Proceeding further we would like to make use of the assigned probabilities for the occurrence of demand and calculate the expected profits for different stocking situations as shown in Table 10.11

Table 10.11: Expected Profit

	Demand						
Stocking Policy	60	61	62	63	64	65	Expected
Probability	0.15	0.20	0.30	0.20	0.10	0.05	Value
60	12	12	12	12	12	12	12
61	11.1	12.2	12.2	12.2	12.2	12.2	12.035
62	10.2	11.3	12.4	12.4	12.4	12.4	11.3
63	9.3	10.4	11.5	12.6	12.6	12.6	11.335
64	8.4	9.5	10.6	11.7	12.8	12.8	10.6
65	7.5	8.6	9.7	10.8	11.9	13.0	9.755

Expected values for;

60 units stocking policy = $12 \times 0.15 + 12 \times 0.2 + 12 \times 0.3 + 12 \times 0.2 + 12 \times 0.1 + 12 \times 0.05$

= 12

61 unit stocking policy = $11.1 \times 0.15 + 12.2 \times 0.2 + 12.2 \times 0.3 + 12.2 \times 0.2 + 12.2 \times 0.1 + 12.2 \times 0.05$

= 12.035 and so on

62 unit stocking policy = 11.3

63 unit stocking policy = 11.335

64 unit stocking policy = 10.6

65 unit stocking policy = 9.7555

From the above situation expected values, Naresh should stock 61 newspapers per day as this stock level gives him the maximum expected value. The next concept we will discuss now is the Expected Value of Perfect Information (EVPI).

EVPI as the name refers is a situation wherein the decision-maker spends money to get additional information so that he gets perfect information for what is going to be the demand on that day so that he can stock only the required quantity and maximize his profits. If Naresh can remove all uncertainty from his problem by obtaining more information on the probability of occurrences of demand he will not be required to incur a loss of Rs. 0.9 on every unsold newspaper. With perfect information available to Naresh the demand will still vary between 60 to 65 units per day. Table 10.12 shows the conditional profit table for Naresh if he has perfect information. It shows the maximum profit that he can expect.

Table 10.12: Conditional profit table

Stocking Policy Probability	Demand						Expected Value
	60	61	62	63	64	65	
	0.15	0.20	0.30	0.20	0.10	0.05	
60	12						1.8
61		12.2					2.44
62			12.4				3.72
63				12.6			2.52
64					12.8		1.28
65						13.0	0.65
						Total	12.41

The maximum profit that Naresh can expect with perfect information is Rs.12.41. In absence of this information, we expected Naresh's profit to be Rs. 12.035. The value of perfect information is Rs. 12.41 – Rs. 12.035 = Rs. 0.375. This means that Naresh should therefore not spend more than Rs. 0.375 every day on gathering perfect information else he will not attain the maximum profit of Rs. 12.41.

Check Your Progress 10.2

Note:

- In Decision making under risk, decision-maker must have _____ knowledge.
- When decisions are at risk, the outcome received is calculated as the expected monetary value.
 - True
 - False
- Decision-making under risk is a situation where more than one state of nature exists and a decision-maker has sufficient information to assign probability values to the likelihood of occurrence of each of these states.
 - True
 - False
- EVPI Stands for
 - Expected Value of Perfect Information (EVPI).
 - Expected Value of Pure Information (EVPI).

- c. Expected Value of Present Information (EVPI).
 - d. Expected Value of Projected Information (EVPI).
5. The probability of demand for cool cab services for hiring on any day in a given district is given in following Table 10.13.

No. of cab demanded	0	1	2	3	4
Probability	0.1	0.2	0.3	0.2	0.2

Cab services have a fixed cost of Rs. 90 each day to keep the daily hire charges (net of variable costs of running) Rs. 200. If the cab services company owns 4 cabs, what is its daily expectation? If the company is about to go into business and currently has no cab how many cabs should it buy?

10.4 DECISION TREE ANALYSIS

The above decision environments deal with decision-making reference to the availability of probabilities or not. Further, both of the environments do not take the time sequence into account in which various actions and consequent events would occur. Decision tree analysis is a technique of decision-making that considers the time sequence.

A tree formation listing of all the possible events and resultant outcomes is presented in a decision tree analysis as we know that decision-making involves several stages and at each stage, each of the alternatives will result in different outcomes and payoff.

The various components of a decision tree are shown in Figure 10.1

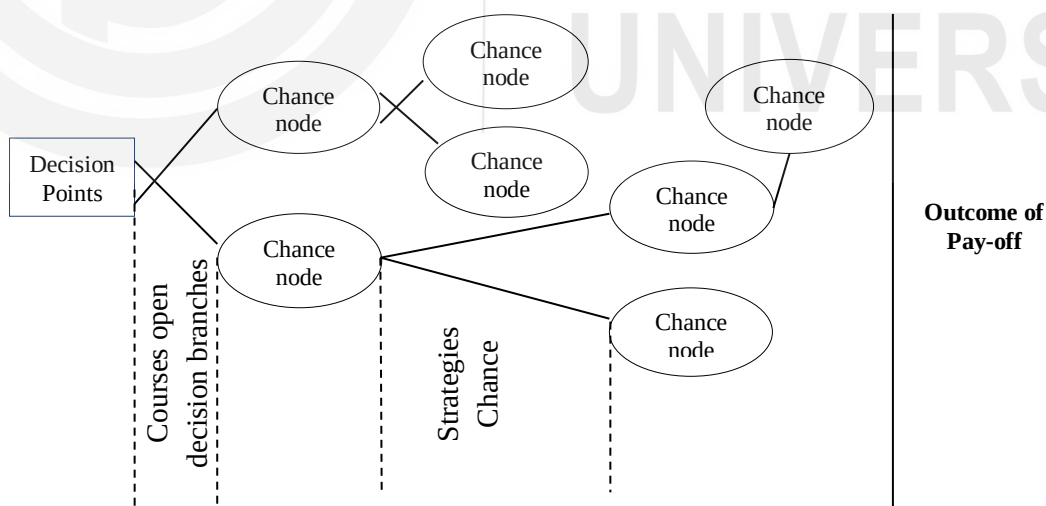


Fig. 10.1: Decision Tree

Decisions Points: The Square indicates a decision point. The branches leading from this square represent various courses of action available to the decision-maker and are called as decision branches.

Node: The node is denoted by a circle. At the end of each branch, there is a node. Various outcomes emerge out of the node with their associated probabilities estimates. Branches also emerge out of the nodes and are called as chance branches. These chance branches are indicative of the likelihood that particular action would be assumed within the given solution. The payoff written in the above figure against each chance branch can be positive or negative depending on the nature of the event. Profits or revenues are represented as positive payoffs whereas expenditure and losses as negative payoffs.

Now let us solve a decision problem using a decision tree with the following illustrative example

Illustrative Example 10.1

1. A farm equipment manufacturing company wants to select between two alternatives as follows:

Alternative I: Distribution through dealers

Alternative II: Direct selling of equipment (owned showroom).

There is a 0.7 probability of having high sales when distributed through a dealer with a payoff of Rs. 70 lakhs. The possible payoff when the sales are low is Rs. 30 lakhs. If the company decides to go for its own distribution (direct-selling) the probability of having high sales is 0.60 with a payoff of Rs. 90 lakhs and Rs. 25 lakhs when sales are low.

Now let us draw the decision tree to understand the problem. We know that there is a choice of two alternatives and hence this can be shown as a decision point in following Figure 10.2.



Fig. 10.2: Decision Tree

Next, we know that there is the possibility of two chances for each alternative, high sales or low sales with some expected payoff. The same is represented as follows in Figure 10.3;

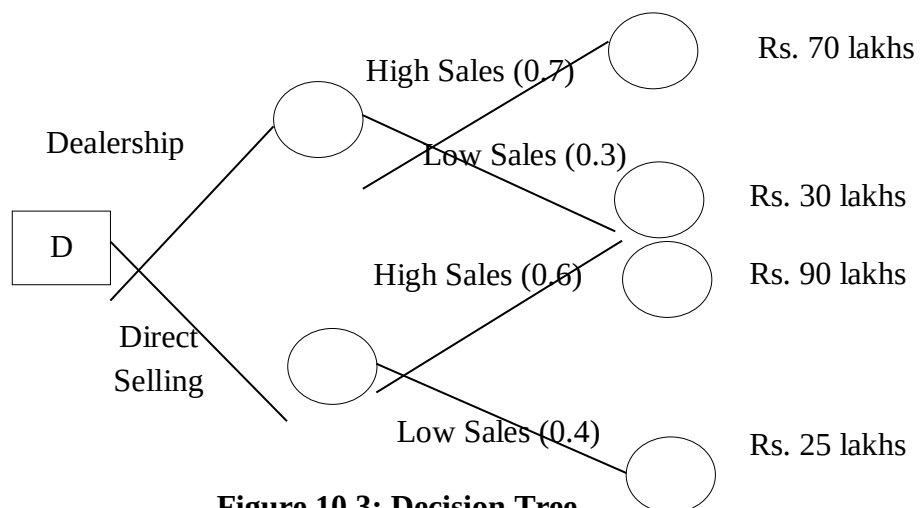


Figure 10.3: Decision Tree

In the above problem, the probability for high sales is given as 0.7 for the dealership, therefore the probability of low sales will be 0.3 ($1 - 0.7 = 0.3$), similarly for direct selling the probability for low sales is calculated as 0.4.

Now we calculate the expected payoffs to decide on the profitable strategy.

Expected payoff for dealership = $(0.7 \times 70) + (0.3 \times 30) = \text{Rs. } 58 \text{ lakhs}$

Expected payoff for direct selling = $(0.6 \times 90) + (0.4 \times 25) = \text{Rs. } 64 \text{ lakhs}$

Based on the above two payoffs it is observed that the direct selling strategy is more attractive with higher expected returns.

So this is how a decision tree can be put into practice. Decision trees can be used to solve complex problems as it offers to present the problem in a graphical representation and can be approached in stages.

Check Your Progress 10.3

1. Decision tree analysis is a technique of decision-making that considers the series sequence.
 - a. True
 - b. False
2. Decision tree analysis uses which types of formation to represent possible events and resultant outcomes.
 - a. Star
 - b. Tree
 - c. Ring

- d. Mesh
3. A decision point is indicated by
 - a. Circle
 - b. Point
 - c. Node
 - d. Square
4. In the decision tree at the end of each branch there is a _____
 - a. Circle
 - b. Point
 - c. Node
 - d. Square
5. In the Decision tree branches that emerge out of the nodes are called as _____ branches.
6. A firm is planning to develop and market a new soft drink. The cost of extensive research to develop the soft drink has been estimated at Rs. 1,00,000. The manager of the research programme has found that there is a 60% chance that the soft drink will be developed successfully. The market potential has been assessed in the following Table 10.14.

Table 10.14: Market Potential

Market Condition	Probability	Present value of profit (Rs.)
Large Market Potential	0.1	50,000
Moderate Market potential	0.6	25,000
Low Market potential	0.3	10,000

The present values figures do not include the cost of research. While the firm is considering this proposal, a second proposal almost similar comes up for consideration. The second one also requires an investment of Rs. 1,00,000 but the present value of all profits is 12,000. The return on investment in the second proposal is certain.

1. Draw a decision tree indicating all events and choices of the firm
2. What decision the firm should take regarding the investment of Rs. 1,00,000?

10.5 LET US SUM UP

Decision theory has become an important technique for decision-making in the face of uncertainty. This unit in detail discussed the decision theory procedures designed to solve problems with a limited number of decision alternatives and a finite list of possible states of nature. The goal of the decision theory approach was to identify the best decision alternatives given an uncertain or risk-filled pattern of future events (i.e., states of nature).

In Decision making under uncertainty, we discussed the application of various decision techniques like maximin, maximax, minimax regret, and the Laplace criterion. When the problems have probabilities known to the decision-maker based on the past data we apply the technique of decision-making under risk and identify the best decision and calculate the expected value of perfect information. The EVPI is the amount the decision maker can pay to get perfect information regarding the occurrence of an event.

Decision trees are used to analyse sequential decision-making situations. The decision alternatives and possible events are structured in the form of a tree diagram. The tree is analysed in a right-to-left manner. Expected value computations are performed at each event node and the optimal decision alternatives are identified at each decision point.

10.6 KEYWORDS

Decision Theory	: It is a method or framework of logical and mathematical concepts, aimed at helping decision-makers to formulate rules and choose among a set of alternatives.
Decision Tree Analysis	: A schematic method of alternatives available to the decision-maker, to analysis the circumstances along with their possible consequences.”
Laplace Criterion	: This method explicitly uses the probability of assessments regarding the likelihood of occurrence of the states of nature.
Maximax Criterion	: This method looks at the best that could happen under each action and then chooses the action with the largest value.
Maximin Criterion	: This method involves selecting the alternative that maximises the minimum pay-off achievable.
Minimax Regrets Criterion	: This method minimizes regret which is highest when one decision has been made instead of another.

13.7 SUGGESTED FURTHER READINGS / REFERENCES

1. Frederick S Hillier and Gerald J Lieberman. (2009), Introduction to Operations Research, Ninth Edition, McGraw-Hill Higher Education.
2. Hira, D. S. and Gupta. P.K. (2007), Operation Research, S Chand and Company, New Delhi.
3. Kalavathy, S. (2002), Operation Research, Fourth Edition, Vikas Publishing House Private Ltd.
4. Panneerselvam, R. (2006), Operation Research, Second Edition, PHI Learning Private Ltd, New Delhi, India.
5. Sharma, A. (2009). Operations Research. Global Media, Himalaya Publishing House.
6. Sharma, J.K. (2010), Operations Research – Problems and Solutions, Third Edition, Macmillan Publishers India Ltd.
7. Shenoy, G.V., Srivastava, U.K. and Sharma, S.C. (2005), Operations Research for Management, Second Edition, New Age International (P) Ltd.
8. Srinivasan, G. (2010), Operations Research – Principles and Applications, Second Edition, PHI Learning Private Ltd, New Delhi, India.
9. Taha, H. A. (2008), Operations Research – An Introduction, Eight Edition, Prentice – Hall of India Private Ltd.

10.8 ANSWER TO CHECK YOUR PROGRESS

Check Your Progress 10.1

1. a
2. a
3. Optimistic
4. Pessimistic
5. False
6. a
7. e
8. c
9. The maximax payoff is Rs. 70,000 corresponding to the alternatives ‘construct’.

Alternatives	States of nature (product demand)				Maximum of row
	High	Moderate	Low	Nil	Rs.

Expand	Rs. 50,000	Rs. 25,000	Rs. – 25,000	Rs. – 25,000	50,000
Contract	Rs. 70,000	Rs. 30,000	Rs. – 40,000	Rs. – 80,000	70,000
Subcontract	Rs. 30,000	Rs. 15,000	Rs. – 1,000	Rs. – 10,000	30,000

← **Maximax**

10.

Type of Biscuits	<i>Profits on estimated level of sales (in lakhs) for quantities</i>		
	5,000	10,000	20,000
Cream (C)	15	25	45
Coconut (Co)	20	55	65
Glucose (G)	25	40	70

Select the minimum payoff for each strategy

Types of Biscuits	<i>Profits on the estimated level of sales (in lakhs) for quantities</i>	
	5,000	
Cream (C)	15	
Coconut (Co)	20	
Glucose (G)	25	

In using the maximin criterion, the decision-maker adopts a pessimistic approach and tries to maximise his security in the face of a highly uncertain situation. For the worst situation, the payoffs are 15, 20, and 25 for C, Co and G respectively. Even at the pessimistic level, the manager tries to make the best of the situation reaching the decision as Glucose, pay-off being the best amongst the worst. This maximises the minimum pay-off. Hence, the company will launch Glucose biscuits (G).

Check Your Progress 10.2

1. Sufficient
2. False
3. True
4. a
5. Fixed Cost = Rs. 90

Variable cost = Rs. 200

Now, we have

No. of cab demanded:	0	1	2	3	4
Pay-offs:	-360	-160	40	240	440

and

$$\text{Daily expectation} = (-360 \times 0.1) + (-160 \times 0.2) + (40 \times 0.3) + (240 \times 0.2) + (440 \times 0.2) = 80$$

Demand of Cabs	Probability	Conditional pay-off				
		0	1	2	3	4
0	0.1	0	-90	-180	-270	-360
1	0.2	0	110	20	-70	-160
2	0.3	0	110	220	130	40
3	0.2	0	110	220	330	240
4	0.2	0	110	220	330	440

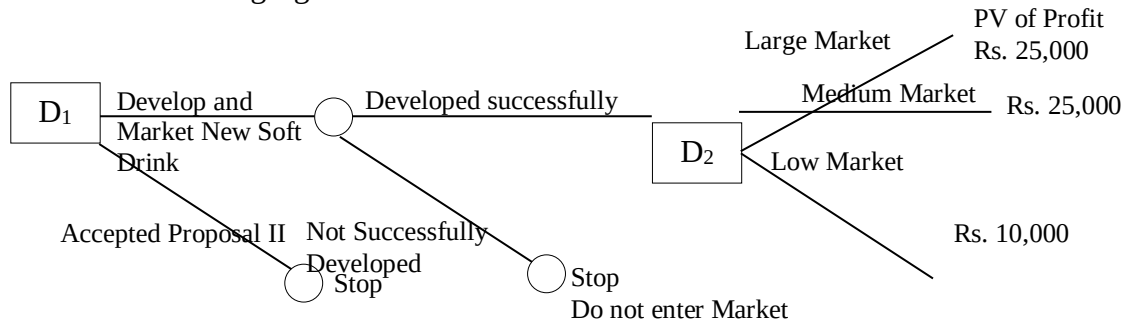
Demand of Cabs	Probability	Conditional pay-off				
		0	1	2	3	4
0	0.1	0	-9	-18	-27	-36
1	0.2	0	22	4	-14	-32
2	0.3	0	33	66	39	12
3	0.2	0	22	44	66	48
4	0.2	0	22	44	66	88
Expected Value		0	90	140	130	80

As the expected value of the course of action is 2 is 140, which is higher than the rest of all, the company should purchase two cabs.

Check Your Progress 10.3

1. False
2. Tree
3. Square
4. Node
5. Chance

6. The following figure is the decision tree



At Point D₂

Decision are (1) Enter the market (2) Do not enter the market

(a) Enter market

EMV = Expected P V

$$= (50,000 \times 0.1) + (25,000 \times 0.6) + (10,000 \times 0.3)$$

$$= 5000 + 15000 + 3000$$

$$= 23,000$$

(b) Do not enter the market

$$\text{EMV} = \text{Expected P V} = 0 \times 1 = 0$$

Decision: Enter the market since EMV is more.

At D₁, Decisions are (1) Develop a new soft drink (2) Accept proposal II

(a) Develop new soft drink

$$\text{EMV} = \text{Expected P V} = (23,000 \times 0.6) + (0 \times 0.4) = (13,800 + 0) = 13,800$$

(b) Accept proposal II

$$\text{EMV} = \text{Expected P V} = 12,000 \times 1 = 12,000$$

Using the EMV criterion, the optimal decision at D₁ is to develop and market the new soft drink.

10.9 UNIT END QUESTIONS

1. What do you understand by the term decision theory?
2. Describe some methods which are useful for decision-making under uncertainty.
3. What techniques are used to solve decision-making problems under uncertainty?
4. Which decision theory techniques result in an optimistic decision? Which technique results in a pessimistic decision?

5. What do you understand by decision-making under risk?
6. Write a note on the value of perfect information.
7. What are the main steps associated with decision tree analysis.

10.10 ASSIGNMENT QUESTIONS

1. Mr. Naresh wants to invest Rs. 10,000 in one of the three options A, B, and C. The payoff for his investment depends on the nature of the benefits he gets from the investment (share market, gold market, fixed deposit). The possible returns under each situation are given below;

Strategy	Nature of Benefits from the investment		
	<i>Share Market (S₁)</i>	<i>Gold Market (S₂)</i>	<i>Fixed Deposit (S₃)</i>
A	2000	1200	1500
B	3000	800	1000
C	2500	1000	1800

What course of action has he to take according to,

- i. Maximin
 - ii. Maximax
 - iii. Laplace
 - iv. Minimax (the regret criterion)
2. At a cake shop, special cookies are sold, it has the following probability of selling cookies.

No. of cookies sold	Probability
10	0.10
11	0.15
12	0.20
13	0.25
14	0.30

The cost of a cookie is 30 paise and the sale price is 50 paise. At end of the day, unsold cookies have to be thrown. How many cookies should the shop owner order to avoid wastage?

3. A glass factory specializing in crystal is developing a substantial backlog and the firm's management is considering three courses of action: Arrange for sub-contracting (S₁) begin overtime production (S₂), and construct new facilities (S₃). The correct choice depends largely upon future demand which may be

low, medium, or high. By consensus, management ranks the respective probabilities as 0.10, 0.50, and 0.40. Cost analysis reveals the effect on the profits that are shown in the table below;

Demand	Probability	Course of Action		
		S ₁ (Subcontracting)	S ₂ (Begin Overtime)	S ₃ (Construct Facilities)
Low (L)	0.10	10	– 20	– 150
Medium (M)	0.50	50	60	20
High (H)	0.40	50	100	200

Show this decision situation in the form of a decision tree and indicate the most preferred decision and corresponding expected value.

10.11 SOLUTION TO ASSIGNMENT QUESTIONS

1. Solution:

i. Maximin criterion

Strategy	Nature of Benefits from the investment			Min	
	(S ₁)	(S ₂)	(S ₃)		
A	2000	1200	1500	1200	Maximin
B	3000	800	1000	800	
C	2500	1000	1800	1000	

Maximum among the minimum pay-offs is 1200. Hence the corresponding option A is to be chosen

ii. Maximax criterion

Strategy	Nature of Benefits from the investment			Max	
	(S ₁)	(S ₂)	(S ₃)		
A	2000	1200	1500	2000	Maximax
B	3000	800	1000	3000	
C	2500	1000	1800	2500	

Maximum among the maximum pay-offs is 3000. Therefore option B is to be chosen.

iii. Laplace criterion

Strategy	Nature of Benefits from the investment			
	(S ₁)	(S ₂)	(S ₃)	
A	2000	1200	1500	$\left(\frac{1}{3} \times 2000\right) + \left(\frac{1}{3} \times 1200\right) + \left(\frac{1}{3} \times 1500\right) = 1566$
B	3000	800	1000	$\left(\frac{1}{3} \times 3000\right) + \left(\frac{1}{3} \times 800\right) + \left(\frac{1}{3} \times 1000\right) = 1600$
C	2500	1000	1800	$\left(\frac{1}{3} \times 2500\right) + \left(\frac{1}{3} \times 1000\right) + \left(\frac{1}{3} \times 1800\right) = 1766$

Alternative C has the maximum expected payoff. Hence alternative C is to be chosen.

iv. Regret criterion

Strategy	Nature of Benefits from the investment			Regret pay-off			Maximum regret
	(S ₁)	(S ₂)	(S ₃)	(S ₁)	(S ₂)	(S ₃)	
A	2000	1200	1500	1000	0	300	1000
B	3000	800	1000	0	400	800	800
C	2500	1000	1800	500	200	0	500
Max	3000	1200	1800				

Among the maximum regrets, we find that 500 is the minimum. Therefore alternative C is to be chosen.

2. We can apply either the EMV criterion or the EOL criterion. Let us apply the EMV criterion for which we have to calculate the payoff. The numbers of cookies ordered are the different courses of action. The cookies ordered maybe 10, 11, 12, 13, 14. Denote them by A₁, A₂, A₃, A₄, A₅.

Similarly, the number of cookies demanded maybe 10, 11, 12, 13 or 14. These demands may be D₁, D₂, D₃, D₄, D₅. These are events. The pay-off values are calculated below;

The selling price of each item = 50 Ps. and cost of a cookie = 30 Ps.

$$A_1D_1 = (10 \times 50) - (10 \times 30) = 200$$

$$A_2D_1 = (10 \times 50) - (11 \times 30) = 170$$

$$A_3D_1 = (10 \times 50) - (12 \times 30) = 140$$

$$A_4D_1 = (10 \times 50) - (13 \times 30) = 110 \text{ etc.}$$

	A₁	A₂	A₃	A₄	A₅
D₁	200	170	140	110	80
D₂	200	220	190	160	130
D₃	200	220	240	210	180
D₄	200	220	240	260	230
D₅	200	220	240	260	280

Given probabilities are 0.10, 0.15, 0.20, 0.25, 0.30

Calculation of EMV for all the Acts

A₁	A₂	A₃	A₄	A₅
Pay off × Prob	Pay off × Prob	Pay off × Prob	Pay off × Prob	Pay off × Prob
$0.10 \times 200 = 20$	$0.10 \times 170 = 17$	$0.10 \times 140 = 14$	$0.10 \times 110 = 11$	$0.10 \times 80 = 8$
$0.15 \times 200 = 30$	$0.15 \times 220 = 33$	$0.15 \times 190 = 28.5$	$0.15 \times 160 = 24$	$0.15 \times 130 = 19.5$
$0.20 \times 200 = 40$	$0.20 \times 220 = 44$	$0.20 \times 240 = 48$	$0.20 \times 210 = 42$	$0.20 \times 180 = 36$
$0.25 \times 200 = 50$	$0.25 \times 220 = 55$	$0.25 \times 240 = 60$	$0.25 \times 260 = 65$	$0.25 \times 230 = 57.5$
$0.30 \times 200 = 60$	$0.30 \times 220 = 66$	$0.30 \times 240 = 72$	$0.30 \times 260 = 78$	$0.30 \times 280 = 84$
= 200	= 215	= 222.5	= 220	= 205

∴ EMV for Acts A₁, A₂, A₃, A₄, and A₅ are respectively 200, 215, 222.5, 220, and 205

EMV for A₃ is greater and therefore A₃ is an optimal act.

∴ No. of cookies to be ordered = 12.

3. A decision tree that represents possible courses of action and states of nature is shown in the following figure. In order to analyze the tree, we start working backward from the end branches.

The most preferred decision at decision node 0 is found by calculating the expected value of each decision branch and selecting the path (course of action) with a high value.

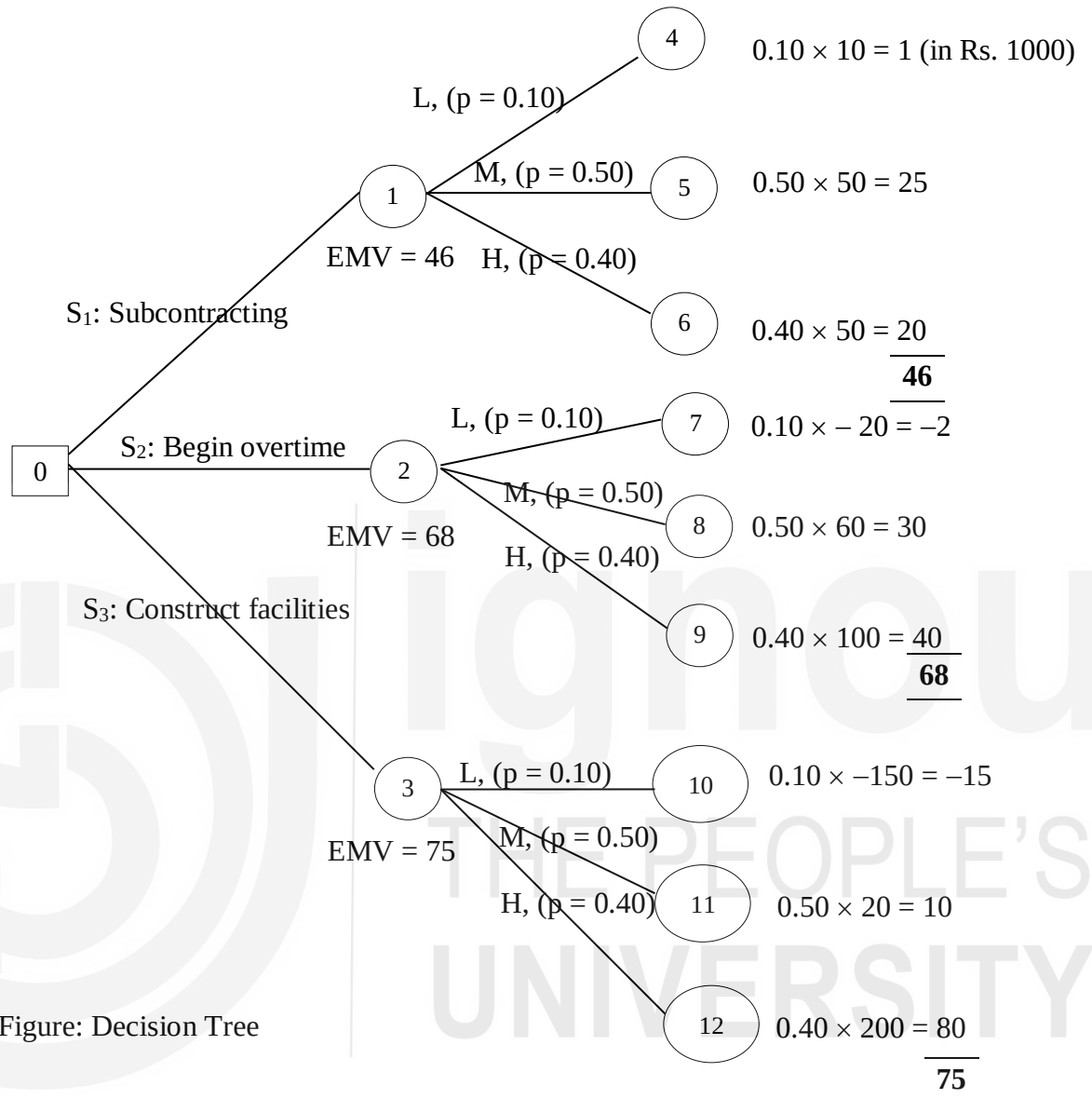


Figure: Decision Tree

Since node 3 has the highest EMV, therefore the decision at node 0 will be to choose the course of action S₃, i.e. construct new facilities.